

Flat Architecture: The Anti-Pattern That Arises When the Three-Level Cell/Aspect/Self Structure per B1.02 Is Not Implemented, Recognizable as All-Cells-No-Aspects, Collapsed Levels, or Missing Self, With Consequences Including Loss of Coordination Governance and Absence of Integration-Level Authority

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

B1.02 commits the CKS architecture to three distinct structural levels — cell, aspect, and Self — each with its own scope, its own governance objects, and its own architectural role. The cell handles a specific informational task. The aspect coordinates cells for a purpose. The Self integrates aspects and their collective intelligence into one unified whole under human governance. These three levels are not optional layers that deployments may or may not implement; they are the structural machinery that makes CKS-governed intelligence coherent at scale.

Flat Architecture is the anti-pattern that arises when this three-level structure is absent or partially instantiated. It takes three recognizable forms: All-Cells-No-Aspects, in which cells operate without any aspect coordination level; Collapsed Levels, in which cell and aspect are conflated into entities whose level cannot be determined; and Missing Self, in which cells and aspects exist but no integrating Self governs their combination. Each form produces distinct operational failures, and all three share a common remediation path: the governed introduction of the missing structural levels through the lifecycle and evolution mechanisms B1.02 is built on.

1. The Commitment Violated: B1.02 Three-Level Structure

B1.02 establishes that CKS-governed intelligence is structured across three levels with distinct architectural scope at each level (§5.2 of the source paper).

The cell is the atomic unit Paper 1 defends — a CKS artifact with substrates, orchestration rules, and the six architectural commitments Paper 1 establishes. All Paper 1 commitments hold at the cell level without modification. The cell handles a specific informational task; its scope is the task it is constituted to perform.

The aspect is a coordination arrangement of cells serving a particular purpose. Multiple aspects can coexist within one Self, potentially sharing cells across arrangements through relational role membership. An aspect operates over its constituent cells as content domain — asking pattern questions, applying coordination rules, governing how cells within the arrangement contribute to its purpose. The aspect's scope is the purpose-defined mode of engagement it constitutes.

The Self is the integrated whole that holds multiple aspects as facets of one CKS-governed intelligence. The Self is not a bigger aspect or a rigid top of a strict access hierarchy; it is what contains the multiple coexisting structural perspectives and their collective intelligence as one unified whole. The Self's scope is integration governance — how aspects combine, how instinct and reasoning are configured at the level of the whole, how cross-aspect conflicts are handled.

The three levels are recursive in architectural commitments: Paper 1's commitments hold at each level. But each level has distinct scope that the others cannot substitute for. A cell cannot perform aspect-level coordination governance, because its scope is a single task, not a purpose-defined arrangement of many tasks. An aspect cannot perform Self-level integration governance, because its scope is one mode of engagement, not the integration of multiple coexisting modes into one whole. And no collection of cells, however large, substitutes for the aspect or Self levels, because the governance functions those levels provide are qualitatively different from the task execution cells perform.

Flat Architecture violates B1.02 by operating with fewer than three functional levels. The violation is architectural: the missing level is not merely absent from a current deployment snapshot but structurally uninstantiated — no entities with the appropriate scope and governance functions exist, and no mechanism is in place to introduce them under governance.

2. Recognizable Forms

Flat Architecture presents in three recognizable sub-forms. Each form has distinct recognition signals derivable from the B2.07–B2.10 decomposition of B1.02.

Form 1: All-Cells-No-Aspects

The deployment consists entirely of cells. No aspect-level coordination structure exists — there are no entities with purpose statements governing cell arrangements, no coordination rules operating over collections of cells for a purpose, and no mechanism through which cells are

coordinated at anything above the individual cell level.

Recognition signals for All-Cells-No-Aspects:

- No entities with purpose statements exist in the substrate (the B2.15 purpose-defined structural arrangement requirement finds nothing). Cells have purposes in the sense of the tasks they perform, but no entity constitutes a coordination arrangement with a purpose that spans multiple cells.
- No coordination rules operating at aspect scope exist (the B2.16 coordination rules requirement finds nothing above cell level).
- No cell carries membership records in any aspect (the B2.08 membership-as-substrate-resident requirement finds no aspect membership records in cell substrates).
- The B2.09 level-distinguishability test for aspect-type entities returns no candidates. Every entity in the deployment passes the cell test and fails the aspect test, because every entity performs a specific informational task and none constitutes a governed coordination arrangement over other entities.

All-Cells-No-Aspects is the most common form of Flat Architecture and the easiest to identify: a substrate inspection that finds only task-performing cells and no coordination-governing aspects is a definitive positive result for this form.

Form 2: Collapsed Levels

Cell and aspect levels are conflated. Entities in the deployment perform both specific informational tasks and coordinate other entities, but these two functions are architecturally indistinguishable in the same entity. The level-distinct scope per B2.07 — which requires that cell-scope entities govern task execution while aspect-scope entities govern cell coordination for a purpose — is absent.

Collapsed Levels is architecturally distinct from All-Cells-No-Aspects in that something resembling aspect-level coordination exists, but it is merged into cell-level entities rather than instantiated as a structurally distinct level. Entities operate at both scopes simultaneously without the architectural separation that B1.02 requires.

Recognition signals for Collapsed Levels:

- The B2.09 level-distinguishability tests are ambiguous rather than clearly negative. Entities partially pass both the cell-type test and the aspect-type test simultaneously — they perform specific informational tasks (cell criterion) but also hold coordination functions over other entities (aspect criterion), without the structural separation that would make the level determination clear.
- The level-instantiation patterns per B2.10 do not cleanly instantiate either a cell-type pattern or an aspect-type pattern. The deployment has a single entity type that handles

both task execution and coordination, and the distinction between these functions is not architecturally governed.

- Membership records per B2.08 are absent or circular: entities may reference each other for coordination purposes, but the substrate-resident membership structure that aspect-type entities should carry is conflated with the task-execution records that cell-type entities carry.

Collapsed Levels often arises when a deployment begins with cells and informally adds coordination responsibilities to individual cells without introducing a structurally distinct aspect level. The result is entities that are locally coherent but architecturally unclassifiable as either cells or aspects.

Form 3: Missing Self

Cells and aspects are present, but no Self integrates them. The deployment has task-performing cells and purpose-coordinating aspects, but lacks the architectural level at which aspects combine into one unified whole under integration governance.

Missing Self is the rarest form of Flat Architecture but has the highest operational consequence, because deployments that reach the aspect level have already handled the complexity of multi-cell coordination and are likely operating at scales where integration governance matters.

Recognition signals for Missing Self:

- No entity carries an integration architecture per B2.21 governing how aspects combine. Each aspect operates as a self-contained coordination arrangement, but no entity governs their combination into a coherent whole.
- No entity carries instinct/reasoning configuration at integration scope per B2.23. The B1.01 instinct/reasoning separation is implemented within aspects and cells, but no entity instantiates that separation at the scope of the whole deployment.
- No entity with cross-aspect conflict handling per B2.21 exists. Conflicts that arise between aspects — where two aspects produce incompatible requirements or contradictory governance decisions — have no governed resolution mechanism at integration scope.
- The B2.09 level-distinguishability test for Self-type entities returns no candidates. Every entity either passes the cell test or the aspect test; none passes the Self test (integration architecture + cross-aspect governance + whole-deployment instinct/reasoning configuration).

3. Emergence Conditions

Flat Architecture does not arise from deliberate architectural choice. Deployments that understand B1.02 and intend to implement it do not produce Flat Architecture. The anti-pattern

emerges from two conditions that are common in the practical lifecycle of CKS-governed deployments.

Incremental development without structural growth. Deployments most naturally begin at cell level. Paper 1's commitments are sufficient for single-cell operation, and early-stage deployments typically instantiate one or a small number of cells handling the most pressing informational tasks. The governance machinery is real and operational at this stage: the substrate carries content, orchestration rules govern cell behavior, humans hold authority over substrate content and rules.

The coordination and integration needs that would motivate aspects and Selves emerge gradually as cell count grows. A deployment that handles each new coordination need through ad-hoc means — informal agreements between cell operators, runtime workarounds, special-case logic added to individual cells — can remain flat indefinitely. The coordination needs are addressed, but they are addressed outside the architecture rather than through it. The result is a growing deployment in which cell-level operation is robust but aspect-level and Self-level governance are structurally absent.

This is the most common path to Flat Architecture: not neglect of the architecture but deferred structural development that never completes. Each incremental step is locally rational; the structural gap accumulates across steps.

Complexity avoidance. Aspect and Self levels introduce architectural obligations. Aspects require purpose statements, coordination rules, membership records, and lifecycle governance. Selves require integration architecture, cross-aspect conflict handling, and instinct/reasoning configuration at integration scope. Architects who are aware of these requirements but find them disproportionate to the current deployment size may deliberately avoid them, keeping the deployment flat to reduce governance overhead.

Complexity avoidance produces well-reasoned Flat Architecture: the architects understand B1.02, assess its requirements as currently disproportionate, and defer instantiation. The result is architecturally identical to incremental-development flatness — the structural levels are absent — but the emergence path differs. Complexity avoidance is more likely to produce Collapsed Levels than All-Cells-No-Aspects, because architects who understand aspects but defer them sometimes informally load aspect coordination responsibilities onto cells rather than leaving them entirely unaddressed.

4. Operational Consequences

Three operational consequences follow from Flat Architecture. Each is a structural consequence, not a contingent failure mode: it follows necessarily from the absence of the missing level.

Ungoverned coordination. Without aspects, cell coordination cannot be governed through coordination rules per B2.16. Whatever coordination occurs between cells happens through ad-hoc means outside the architecture: informal agreements between cell operators, special-case logic within cells, runtime workarounds. This coordination is not inspectable as substrate content, not modifiable under governance, and not overridable through the human authority architecture Paper 1 establishes. The Paper 1 commitments — human-governed, AI-as-substrate-mediator, path retraceability, conflict preservation — apply to cell-level operation but not to the inter-cell coordination that occurs outside the architecture. Coordination decisions that should be substrate-resident governance objects become invisible to the governance machinery.

Absent integration governance. Without Self, the integration of multiple operational capabilities cannot be governed as an architectural object. Aspects — where they exist — operate as independent coordination arrangements without a level at which their combination is governed. Cross-aspect conflicts have no substrate-resident resolution mechanism. Instinct/reasoning configuration at integration scope has no architectural home. The deployment may exhibit integrated behavior at runtime — aspects may happen to compose coherently in practice — but this coherence is not architecturally governed. It is a contingent runtime property that can silently change when aspects evolve independently, when new aspects are introduced, or when conflicts between aspects emerge that the ad-hoc coordination mechanisms are not equipped to handle.

Scaling failure. Flat Architecture fails to scale coherently as deployment scope grows. Each new cell added to an All-Cells-No-Aspects deployment increases the ungoverned coordination surface proportionally — more cells mean more inter-cell coordination that must be handled outside the architecture. Each new aspect added to a Missing-Self deployment increases the ungoverned integration surface proportionally — more aspects mean more cross-aspect interactions that the absent Self cannot govern. The scaling failure is not a performance problem but a governance-capacity problem: the ungoverned surface grows while the governed surface remains fixed at cell level. At small scales this gap is manageable; at the scales where Paper 2's architecture becomes valuable — multi-aspect enterprise deployments — it is not.

5. Detection

Two detection procedures identify Flat Architecture. Both derive from the B2.09 level-distinguishability tests and the B2.10 level-instantiation patterns that the B1.02 decomposition establishes.

Procedure 1: B2.09 level-distinguishability test. Apply the B2.09 level-distinguishability tests to the deployment's full entity population:

- The *cell-type test* checks whether an entity performs a specific informational task under its own orchestration rules, carries DNA and action layers, and holds Paper 1

commitments at cell scope.

- The *aspect-type test* checks whether an entity holds a purpose statement governing a cell arrangement, carries coordination rules over constituent cells, and holds aspect-scope membership records per B2.08.
- The *Self-type test* checks whether an entity carries integration architecture per B2.21, instinct/reasoning configuration at integration scope per B2.23, and cross-aspect conflict governance.

A deployment in which the aspect-type test returns no passing entities has All-Cells-No-Aspects. A deployment in which the aspect-type and cell-type tests return ambiguous results — entities that partially pass both — has Collapsed Levels. A deployment in which the Self-type test returns no passing entities has Missing Self. Any deployment with fewer than three levels instantiated, as determined by this procedure, has Flat Architecture in one or more sub-forms.

Procedure 2: B2.10 level-instantiation patterns verification. Independent of the per-entity distinguishability test, verify that the deployment instantiates the structural patterns each level requires. Per B2.10:

- Cell-level instantiation is confirmed when the deployment contains substrate-resident cells with DNA layers, action layers, orchestration rules, and Paper 1 commitments.
- Aspect-level instantiation is confirmed when the deployment contains substrate-resident aspect entities with purpose statements, coordination rules over constituent cells, and membership records per B2.08.
- Self-level instantiation is confirmed when the deployment contains a substrate-resident Self entity with integration architecture, instinct/reasoning configuration at integration scope, and cross-aspect conflict handling.

Absence of aspect-level or Self-level instantiation patterns, confirmed through substrate inspection, constitutes a positive detection result for Flat Architecture.

6. Remediation

Flat Architecture is remediable through the governed introduction of the missing structural levels. Remediation does not require redesigning existing cell-level architecture; the cells and their Paper 1 commitments are sound. Remediation requires adding the structural levels above the cell level that B1.02 specifies.

Step 1: Introduce aspect-level coordination structure. For All-Cells-No-Aspects and Collapsed Levels, introduce aspect entities through governed birth per B1.06. Each new aspect entity is instantiated with a purpose statement identifying the cell arrangement it coordinates, coordination rules governing how it operates over constituent cells, and membership records per B2.08 establishing which cells participate in the aspect and under what governance

properties.

Where Collapsed Levels is the presenting form, the remediation additionally requires structural separation of the coordination functions that were conflated into cell-level entities. Coordination rules that were informally embedded in cells are extracted and formalized as aspect-level substrate content, making them inspectable, modifiable, and overridable through the human authority architecture.

Cell coordination logic that was previously handled through ad-hoc means is migrated to aspect coordination rules per B2.16. This migration makes previously invisible coordination decisions substrate-resident governance objects, restoring the Paper 1 commitments to coordination operations that were outside their scope under Flat Architecture.

Step 2: Introduce Self-level integration. Once aspects are instantiated, introduce the Self entity through governed birth per B1.06. The Self is instantiated with integration architecture per B2.21 governing how aspects combine, instinct/reasoning configuration at integration scope per B2.23, and cross-aspect conflict handling mechanisms.

For Missing Self — where aspects already exist — this step does not require changes to the existing aspect structure. It adds the integration governance level that the existing aspects lack.

Step 3: Govern the transition through directed selection. The introduction of new structural levels is not a one-time architectural change but a structural evolution of the deployment. Managed through directed selection per B1.14, the transition proceeds incrementally: aspects are introduced for the most coordination-critical cell arrangements first; Self-level integration is introduced when the aspect population reaches the scale at which integration governance produces meaningful governance benefit. At each step, the introduction is governed — the new entities are born through governed birth, their substrate content is human-authored and human-governed, and the transition is traceable through the path retraceability commitment Paper 1 establishes.

7. Operational Test

A deployment is free of Flat Architecture if and only if all of the following are true:

1. The B2.09 level-distinguishability test identifies at least one entity that unambiguously passes the cell-type test and at least one that unambiguously passes the aspect-type test.
2. The B2.09 level-distinguishability test identifies at least one entity that unambiguously passes the Self-type test.
3. The B2.10 level-instantiation patterns verification confirms substrate-resident aspect entities with purpose statements, coordination rules, and membership records per B2.08.
4. The B2.10 level-instantiation patterns verification confirms a substrate-resident Self entity with integration architecture per B2.21 and instinct/reasoning configuration at

integration scope per B2.23.

5. No entity simultaneously and ambiguously passes both the cell-type and aspect-type tests (no Collapsed Levels).

6. Cell coordination that spans more than one cell is governed by aspect coordination rules per B2.16, not by ad-hoc means outside the substrate.

7. Cross-aspect coordination is governed by Self-level integration architecture per B2.21, not by ad-hoc means outside the substrate.

A deployment that fails any of (1)–(7) has Flat Architecture in one or more sub-forms. The specific sub-form is identified by which items fail: failure of (1) and (3) and (6) indicates All-Cells-No-Aspects; failure of (5) indicates Collapsed Levels; failure of (2) and (4) and (7) indicates Missing Self.

Source Papers

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

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